



PPG INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

(Approved by AICTE, New Delhi & Affiliated to Anna University, Chennai)
Recognized by UGC | Accredited by NAAC with "A" | ISO Certified Institution
NH 209, Sathy Main Road, Saravanampatti, Coimbatore – 641035, Tamil Nadu.



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CCS354 - NETWORK SECURITY



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CCS354 – NETWORK SECURITY

Name: _____ Roll No: _____

Semester: _____ Branch: _____

Certified Bonafide Record of work done by _____

Place: Coimbatore

Date:

Staff In-Charge

Head of the Department

University Register Number: _____

Submitted for the University Practical Examination held on: _____

Internal Examiner

External Examiner

LIST OF EXPERIMENTS

Ex.no	Date	Name of the Experiment	Page No	Marks	Sign
1		Implementing symmetric key algorithm -DES			
2		Implementing Key exchange algorithm			
3		Implement the SIGNATURE SCHEME – Digital Signature standard			
4		Analysis of TCP/UDP Traffic Using Wireshark and Tcpdump			
5		Check message integrity and confidentiality using SSL			
6		Experiment Eavesdropping, Dictionary attacks, MITM attacks			
7		Experiment with Sniff Traffic using ARO Poisoning			
8		Demonstrate intrusion detection system using any tool			
9		Explore network monitoring tools			
10		Study to configure Firewall,VPN			

Ex no: 01	Data Encryption Standard (DES) Algorithm (User Message Encryption)
Date :	

AIM:

To use Data Encryption Standard (DES) Algorithm for a practical application like User Message Encryption

ALGORITHM:

1. Create a DES Key
2. Create a Cipher instance from Cipher class, specify the following information and separated by a slash(/)
 - a. Algorithm name
 - b. Mode (optional)
 - c. Padding scheme (optional)
3. Convert String into Byte[] array format.
4. Make Cipher in encrypt mode, and encrypt it with Cipher.doFinal() method.
5. Make Cipher in decrypt mode, and decrypt it with Cipher.doFinal() method.

PROGRAM:

DES.java

```
import javax.crypto.Cipher;
import javax.crypto.KeyGenerator;
import javax.crypto.SecretKey;

public class DESExample {
    public static void main(String[] args) {
        try {
            System.out.println("Message Encryption Using DES Algorithm\n");
            KeyGenerator keyGenerator = KeyGenerator.getInstance("DES");
            SecretKey secretKey = keyGenerator.generateKey();
            Cipher cipher = Cipher.getInstance("DES/ECB/PKCS5Padding");
            String message = "Secret Information";
```

```

        byte[] text = message.getBytes();
        System.out.println("Original Message: " + message);
        cipher.init(Cipher.ENCRYPT_MODE, secretKey);
        byte[] encryptedText = cipher.doFinal(text);
        System.out.println("Encrypted Message: " + encryptedText);
        cipher.init(Cipher.DECRYPT_MODE, secretKey);
        byte[] decryptedText = cipher.doFinal(encryptedText);
        System.out.println("Decrypted Message: " + new String(decryptedText));
    } catch (Exception e) {
        e.printStackTrace();
    }
}
}

```

OUTPUT:

Message Encryption Using DES Algorithm

Original Message: Secret Information

Encrypted Message: [B@6d06d69c

Decrypted Message: Secret Information

RESULT:

Thus the java program for DES Algorithm has been implemented and the output verified successfully.

Ex no: 02	Key Exchange Algorithm
Date :	

AIM:

To implement the Diffie-Hellman Key Exchange algorithm for a given problem.

ALGORITHM:

1. Alice and Bob publicly agree to use a modulus $p=23$ and base $g=5$.
2. Alice chooses a secret integer $a=4$. Then sends Bob $A=g^a \bmod p$.
3. Bob chooses a secret integer $b=3$. Then sends Bob $B=g^b \bmod p$.
4. Alice computes $s=B^a \bmod p$.
5. Bob computes $s=A^b \bmod p$.
6. Alice and Bob now share a secret.

PROGRAM:

```
public class DiffieHellman {  
    public static void main(String[] args) {  
        int p = 23; // Prime number  
        int g = 5; // Primitive root  
        int a = 4; // Alice secret  
        int b = 3; // Bob secret  
        int A = (int) (Math.pow(g, a) % p);  
        int B = (int) (Math.pow(g, b) % p);  
        int aliceSecret = (int) (Math.pow(B, a) % p);  
        int bobSecret = (int) (Math.pow(A, b) % p);  
        System.out.println("Simulation of Diffie-Hellman Key Exchange\n");  
        System.out.println("Alice sends: " + A);  
        System.out.println("Bob sends: " + B);  
        System.out.println("Alice computes secret: " + aliceSecret);  
        System.out.println("Bob computes secret: " + bobSecret);  
    }  
}
```

```
if (aliceSecret == bobSecret) {  
    System.out.println("Shared Secret Key: " + aliceSecret);  
} else {  
    System.out.println("Error: Keys do not match");  
}  
}  
}
```

OUTPUT:

Simulation of Diffie-Hellman Key Exchange

Alice sends: 4

Bob sends: 10

Alice computes secret: 18

Bob computes secret: 18

Shared Secret Key: 18

RESULT:

Thus the Diffie Hellman key exchange algorithm has been implemented using Java Program and the output has been verified successfully.

Ex no: 03	Digital Signature Standard
Date :	

AIM:

To implement the SIGNATURE SCHEME Digital Signature Standard.

ALGORITHM:

1. Create a KeyPairGenerator object.
2. Initialize the KeyPairGenerator object.
3. Generate the KeyPairGenerator...
4. Get the private key form the pair.
5. Create a signature object.
6. Initialize the Signature object.
7. Add data to the Signature object.
8. Calculate the Signature.

PROGRAM:

```
import java.security.*;
import java.util.Scanner;

public class DigitalSignatureExample {

    public static void main(String[] args) throws Exception {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter message: ");

        String message = sc.nextLine();

        KeyPairGenerator keyPairGen = KeyPairGenerator.getInstance("DSA");

        keyPairGen.initialize(2048);

        KeyPair pair = keyPairGen.generateKeyPair();

        PrivateKey privateKey = pair.getPrivate();

        Signature signature = Signature.getInstance("SHA256withDSA");

        signature.initSign(privateKey);
```



```
signature.update(message.getBytes());  
byte[] digitalSignature = signature.sign();  
System.out.println("Digital Signature:");  
for (byte b : digitalSignature) {  
    System.out.print(String.format("%02X", b));  
}  
}  
}
```

OUTPUT:

Enter message: Hello World

Digital Signature:

3046022100A1B2C3D4E5F67890ABCDEF1234567890ABCDEF1234567890ABCD
EF1234567890022100F1E2D3C4B5A697887766554433221100FFEEDDCCBBAA9
9887766554433221100

RESULT:

Thus the Digital Signature Standard Scheme has been implemented and the output has been verified successfully.

Ex no: 04	Analysis of TCP/UDP Traffic Using Wireshark and Tcpdump
Date :	

AIM:

Installation of Wire shark, tcpdump and observe data transferred in client server communication using UDP/TCP and identify the UDP/TCO datagram.

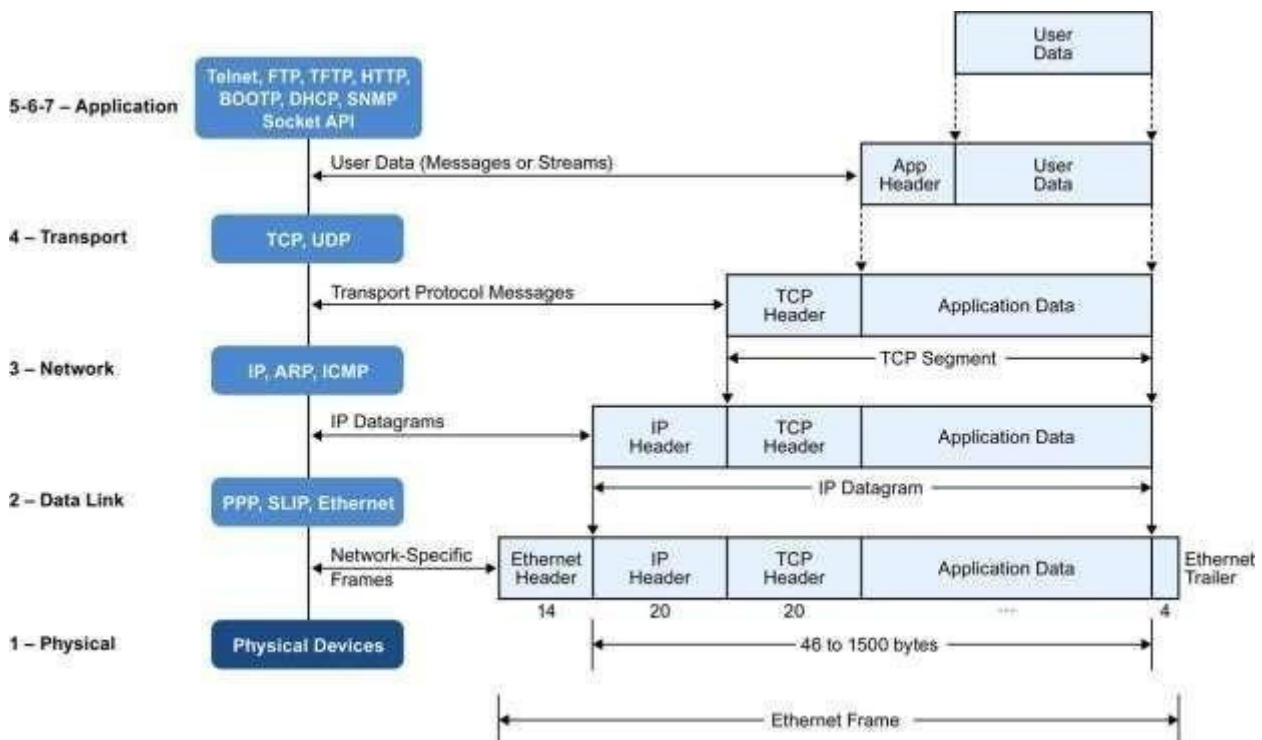
INTRODUCTION:

The first part of the lab introduces packet sniffer, Wireshark. Wireshark is a free open source network protocol analyser. It is used for network troubleshooting and communication protocol analysi. Wireshark captures network packets in real time and display them in human readable format. It provides many advanced features including live capture and offline analysis, three pane packet browser, colouring rules for analysis. This document uses Wireshark for the experiments, and it covers Wireshark installation, packet capturing, and protocol analysis.



BACKGROUND

TCP/IP Network Stack

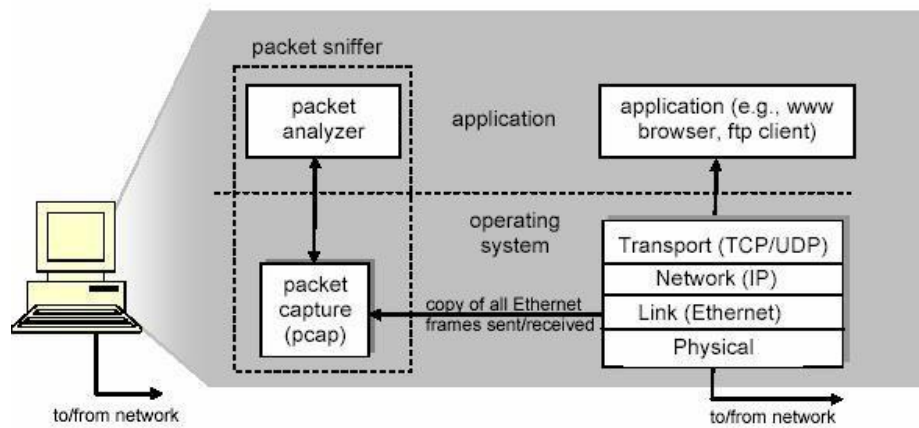


In the CSC 4190 Introduction to Computer Networking (one of the prerequisite courses), TCP/IP network stack is introduced and studied. This background section briefly explains the concept of TCP/IP network stack to help you better understand the experiments. TCP/IP is the most commonly used network model for Internet services. Because its most important protocols, the Transmission Control Protocol (TCP) and the Internet Protocol (IP) were the first networking protocols defined in this standard, it is named as TCP/IP. However, it contains multiple layers including application layer, transport layer, network layer, and data link layer

Packet Sniffer

Packet sniffer is a basic tool for observing network packet exchange in a computer. As the name suggests, a packet sniffer captures (“sniffs”) packets being sent/received from/ by your computer.

The second component of a packet sniffer is the packet analyser, which displays the contents of all fields within a protocol message. In order to do so, the packet analyser,



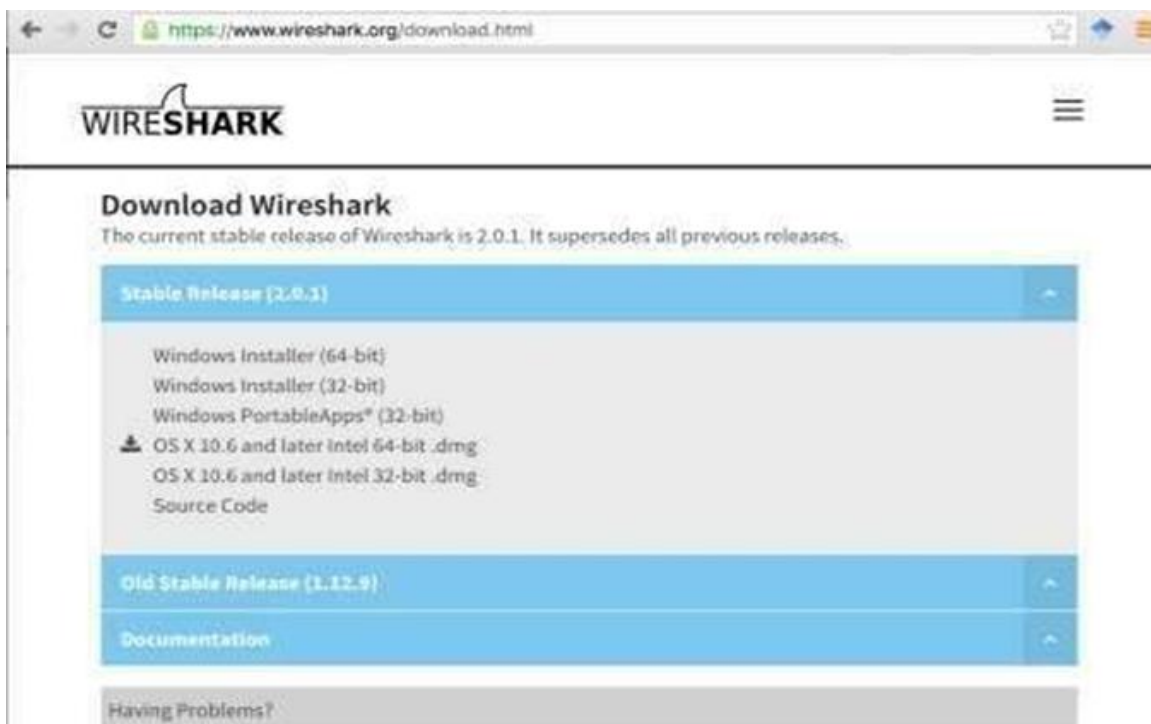
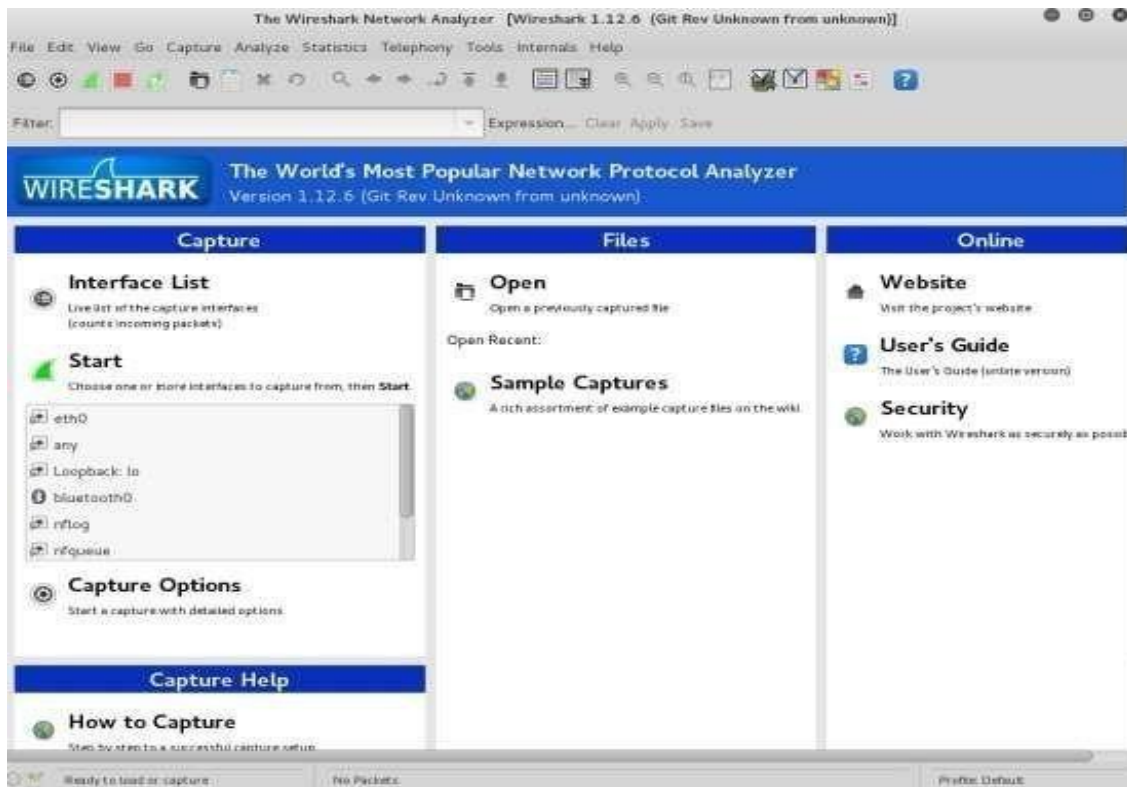
For example, suppose we are interested in displaying the various fields in messages exchanged by the HTTP protocol in **Figure 3**. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string “GET,” “POST,” or “HEAD”.

We will be using the Wireshark packet sniffer [<http://www.wireshark.org/>] for these labs, allowing us to display the contents of messages being sent/received from/by protocols at different levels of the protocol stack. (Technically speaking, Wireshark is a packet analyzer that uses a packet capture library in your computer). Wireshark is a free network protocol analyzer that runs on Windows, Linux/Unix, and Mac computers.

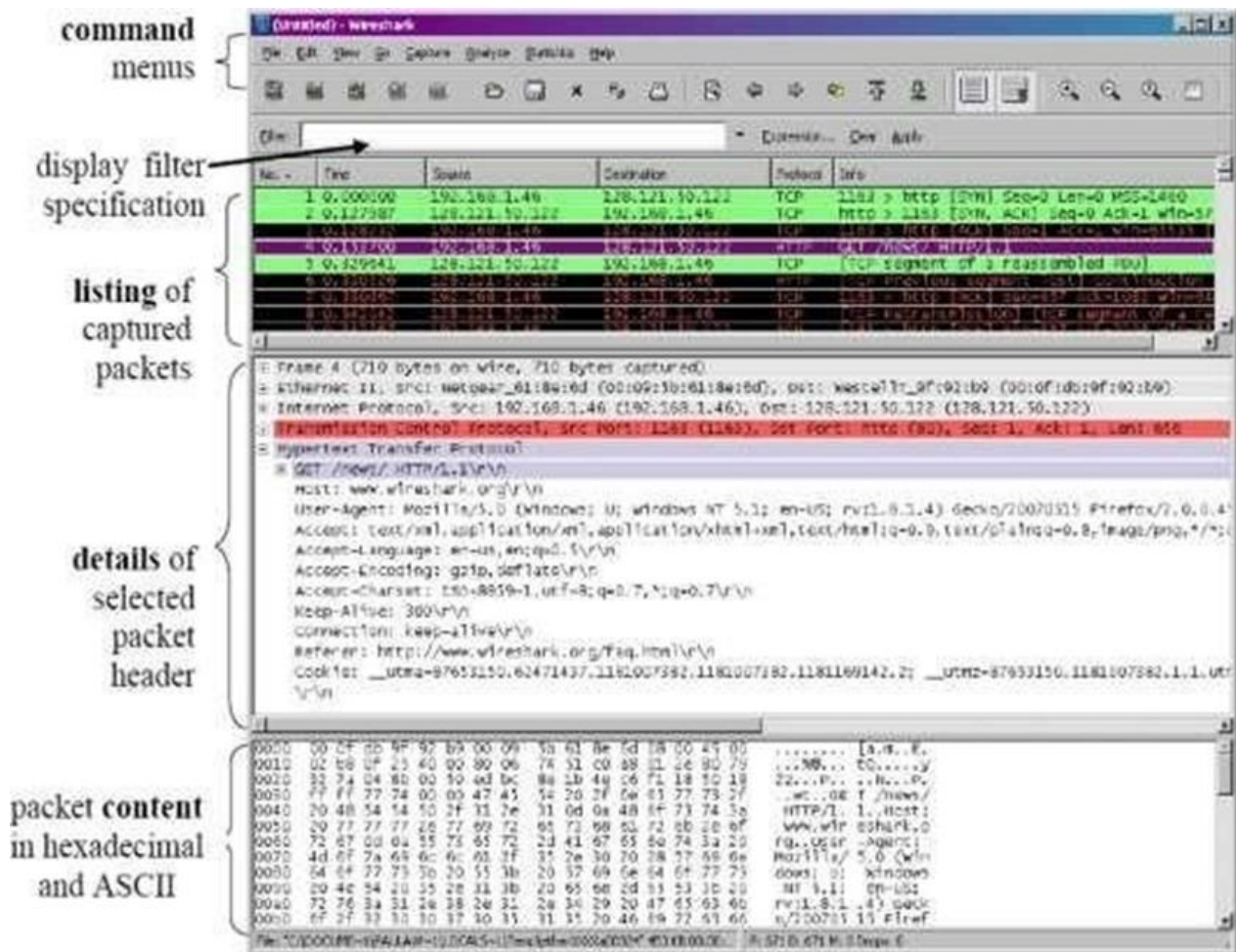
Getting Wireshark

The Kai Linux has Wireshark installed. You can just launch the Kali Linux VM and open Wireshark there. Wireshark can also be downloaded from here:

<https://www.wireshark.org/download.html>



STARTING WIRESHARK:



The Wireshark interface has five major components:

The **command menus** are standard pulldown menus located at the top of the window. Of interest to us now is the File and Capture menus. The File menu allows you to save captured packet data or open a file containing previously captured packet data, and exit the Wireshark application. The Capture menu allows you to begin packet capture.

The **packet-listing window** displays a one-line summary for each packet captured, including the packet number (assigned by Wireshark; this is not a packet number contained in any protocol's header), the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet. The packet listing can be sorted according to any of these categories by

clicking on a column name. The protocol type field lists the highest- level protocol that sent or received this packet, i.e., the protocol that is the source or ultimate sink for this packet.

The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. (To select a packet in the packet-listing window, place the cursor over the packet's one- line summary in the packet-listing window and click with the left mouse button.). The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

Towards the top of the Wireshark graphical user interface, is the **packet display filter field**, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window (and hence the packet-header and packet-contents windows). In the example below, we'll use the packet-display filter field to have Wireshark hide (not display) packets except those that correspond to HTTP messages.

Capturing Packets

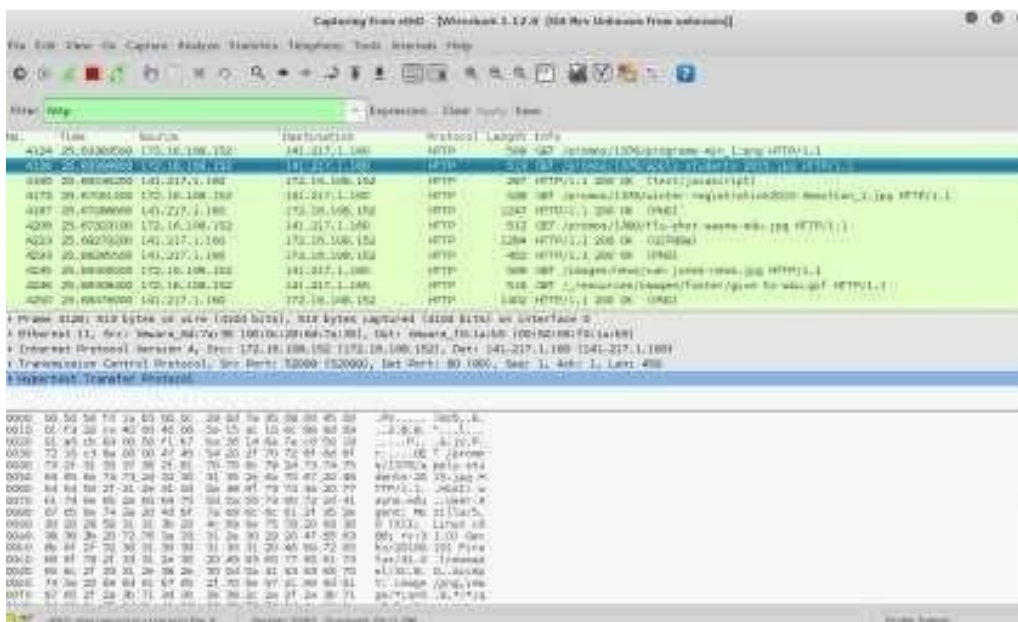
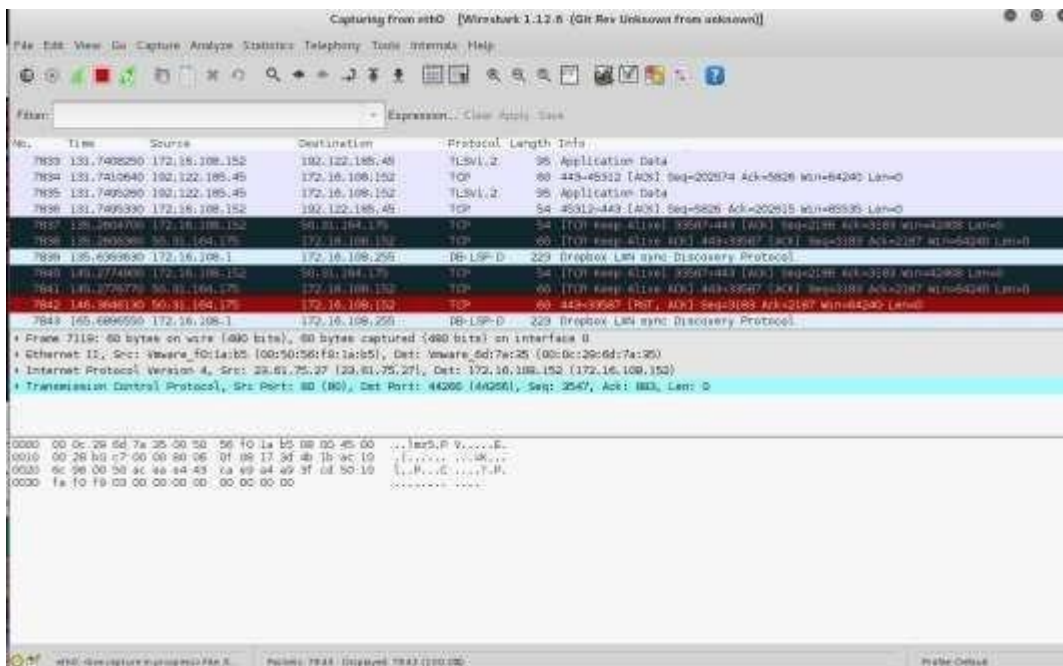
After downloading and installing Wireshark, you can launch it and click the name of an interface under Interface List to start capturing packets on that interface. For example, if you want to capture traffic on the wireless network, click your wireless interface.

Test Run

Do the following steps:

1. Start up the Wireshark program (select an interface and press start to capture packets).
2. Start up your favorite browser (ceweasel in Kali Linux).
3. In your browser, go to Wayne State homepage by typing www.wayne.edu.
4. After your browser has displayed the <http://www.wayne.edu> page, stop Wireshark packet capture by selecting stop in the Wireshark capture window. This will cause the Wireshark capture window to disappear and the main

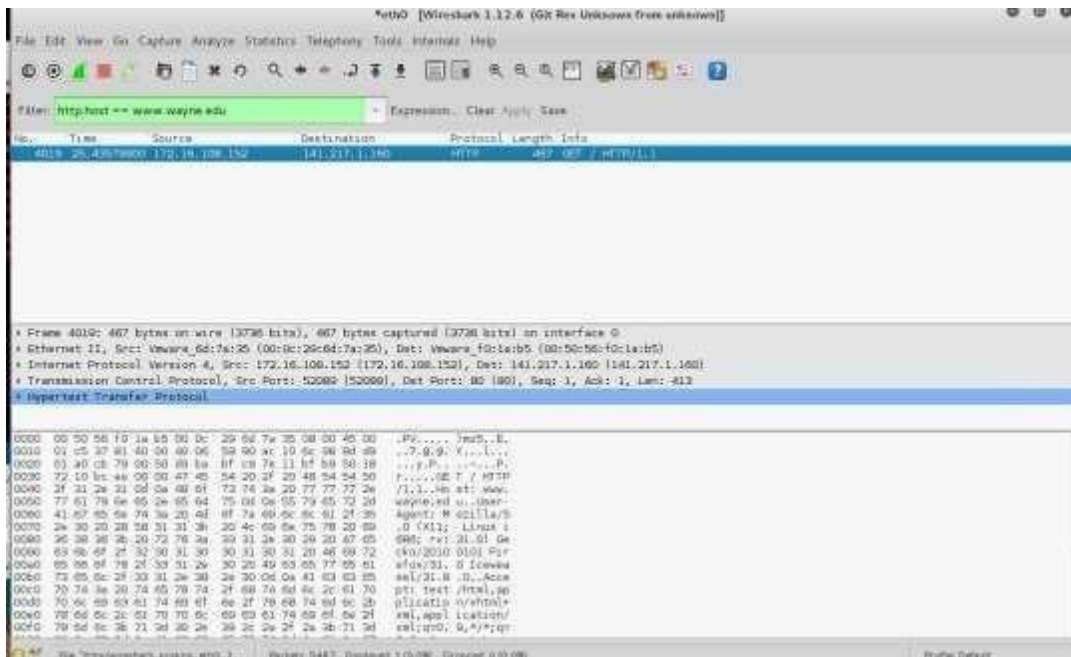
Wireshark window to display all packets captured since you began packet capture see image below:



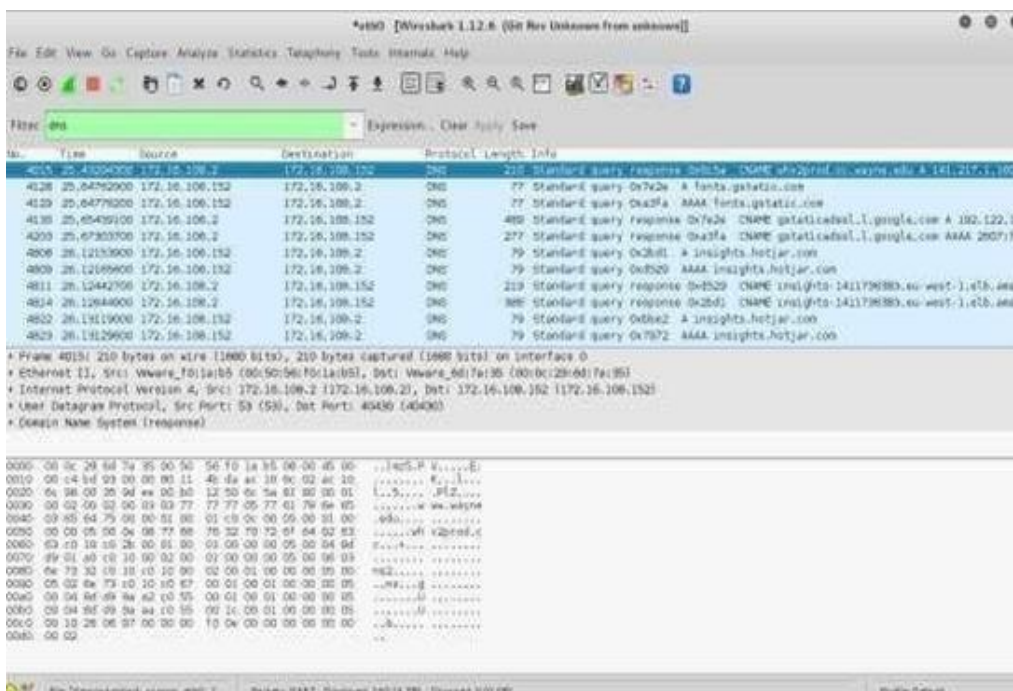
5. Color Coding : You'll probably see packets highlighted in green, blue, and black. Wireshark uses colors to help you identify the types of traffic at a glance. By default, green is TCP traffic, dark blue is DNS traffic, light blue is UDP traffic, and black identifies TCP packets with problems for example, they could have been delivered out of order.

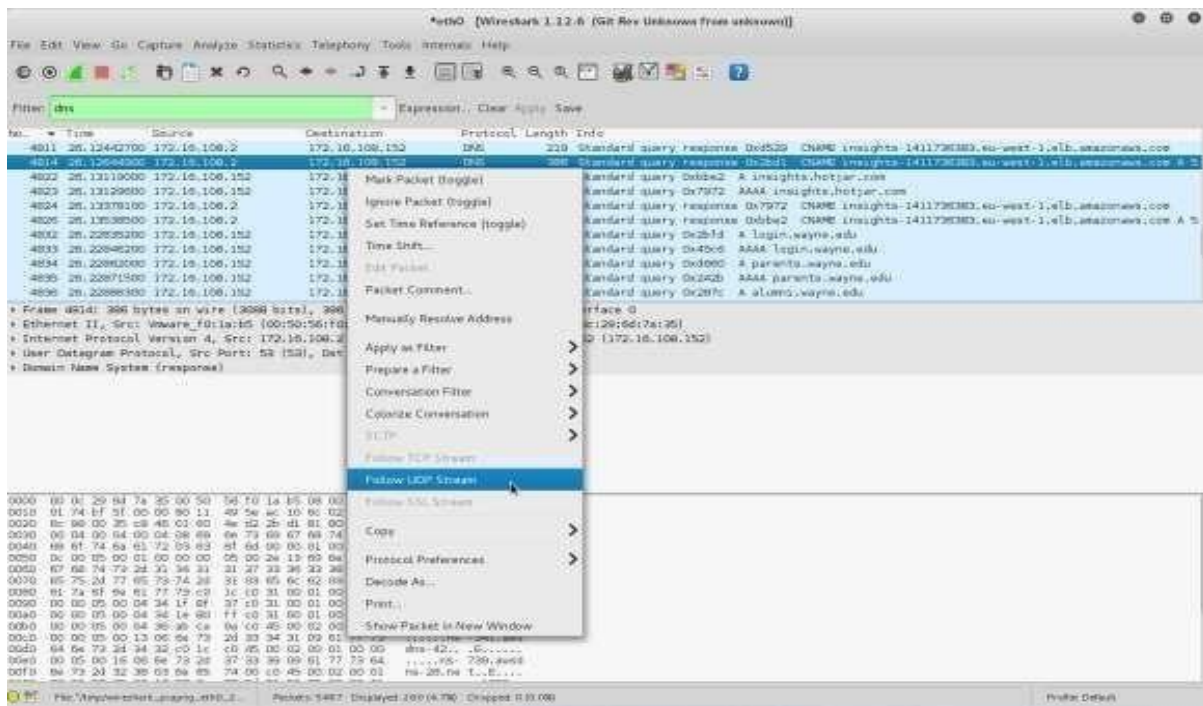
6. You now have live packet data that contains all protocol messages exchanged between your computer and other network entities! However, as you will notice the

HTTP messages are not clearly shown because there are many other packets included in the packet capture. Even though the only action you took was to open your browser, there are many other programs in your computer that communicate via the network in the background. To filter the connections to the ones we want to focus on, we have to use the filtering functionality of Wireshark by typing “http” in the filtering field as shown below:



7. Now, we can try another protocol. Let's use DNS protocol as an example here.





8. IF we close this window and change the filter back to “http.host www.wayne.edu” and then follow a packet from the list of packets that match that filter, we should get the something similar to the following screens. Note that we click on Follow TCP Stream this time.



RESULT:

Installation of Wire shark, tcpdump to observe data transferred in client server communication using UDP/TCP and identify the UDP/TCP datagram is successfully completed.

Ex no: 05	Check Message Integrity and Confidentiality using SSL
Date :	

AIM:

To calculate the message digest of a text using the SHA-1 algorithm.

ALGORITHM

1. Append padding bits
2. Append Length -64its are appended to the end.
3. Prepare processing functions
4. Prepare processing constants
5. Initialize buffers
6. Processing message in 512 bit blocks.

PROGRAM

Sha1.java

```
import java.security.MessageDigest;

public class sha1 {

    public static void main(String[] args) {

        try {

            MessageDigest md = MessageDigest.getInstance("SHA-1");

            System.out.println("Message digest object info:\n");

            System.out.println("Algorithm = " + md.getAlgorithm());

            System.out.println("Provider = " + md.getProvider());

            System.out.println("ToString = " + md.toString());

            String input = "";

            md.update(input.getBytes());

            byte[] output = md.digest();

            System.out.println("\nSHA1(\"" + input + "\") = " + bytesToHex(output));
```

```

        input = "abc";
        md.update(input.getBytes());
        output = md.digest();
        System.out.println("\nSHA1(\"" + input + "\") = " + bytesToHex(output));
        input = "abcdefghijklmnopqrstuvwxy";
        md.update(input.getBytes());
        output = md.digest();
        System.out.println("\nSHA1(\"" + input + "\") = " + bytesToHex(output));
    } catch (Exception e) {
        System.out.println("Exception: " + e);
    }
}

private static String bytesToHex(byte[] b) {
    char[] hexDigit = {
        '0','1','2','3','4','5','6','7',
        '8','9','A','B','C','D','E','F'
    };
    StringBuilder buf = new StringBuilder();
    for (byte aB : b) {
        buf.append(hexDigit[(aB >> 4) & 0x0F]);
        buf.append(hexDigit[aB & 0x0F]);
    }
    return buf.toString();
}
}

```

OUTPUT

Message digest object info:

Algorithm = SHA-1

Provider = SUN version 1.8

ToString = SHA1 Message Digest from SUN, <initialized>

SHA1("") = DA39A3EE5E6B4B0D3255BFEF95601890AFD80709

SHA1("abc") = A9993E364706816ABA3E25717850C26C9CD0D89D

SHA1("abcdefghijklmnopqrstuvwxyz") =
32D10C7B8CF96570CA04CE37F2A19D84240D3A89

RESULT

Thus the Secure Hash Algorithm (SHA-1) has been implemented and the output has been verified successfully.

Ex no: 06

Date :

Eavesdropping, Dictionary attacks, MITM attacks

AIM

To experiment eavesdropping, dictionary attacks, MITM attacks.

VISUAL OBJECTIVE



INTRODUCTION

Password cracking is a term used to describe the penetration of a network, system, or resource with or without the use of tools to unlock a resource that has been secured with a password. Password cracking tools may seem like powerful decryptors, but in reality are little more than fast, sophisticated guessing machines.

TYPES OF PASSWORD BREAKING

1. Dictionary attack

- a. A simple dictionary attack is usually the fastest way to break into a machine. A dictionary file is loaded into a cracking application, which is run against user accounts located by the application.

2. Brute force attack

- a. A brute force attack is a very powerful form of attack, though it may often take a long time to work depending on the complexity of the password. The program will begin trying any and every combination of numbers and letters and running them against the hashed passwords.

3. Hybrid attack

- a. Another well known form of attack is the hybrid attack. A hybrid attack will add numbers or symbols to the search words to successfully crack a

password. Many people change their passwords by simply adding a number to the end of their current password.

CRACKING PROCESS

Since a brute force attack is the most time consuming and is not likely to break any passwords that are not composed of random characters, the best plan is to use techniques that are computationally efficient compared to untargeted and unspecific techniques. By applying what is known about how users select passwords, an intruder can tremendously increase the odds in their favor of finding passwords. With the right techniques, some poor passwords can be cracked in under a second.

1. Alternate upper and lowercase lettering.
2. Spell the word forward and then backward, and then fuse the two results.
3. Add the number 1 to the beginning and or end of each word.

Naturally, the more rules one applies to the words, the longer the cracking process takes. However, more rules also guarantee a higher likelihood of success.

TASK 1- MICROSOFT OFFICE PASSWORD RECOVERY

Many applications require you to establish an ID and password that may be saved and automatically substituted for future authentication. The password will usually appear on the screen as a series of asterisks. This is fine as long as your system remembers the password for you but what if it "forgets" or you need it for use on another system. Fortunately, many utilities have been written to recover such passwords. In this task, you will use OfficeKey to recover the password for a MS word document.

Step 1: Find the folder “Lab1” on your desktop, and open it.

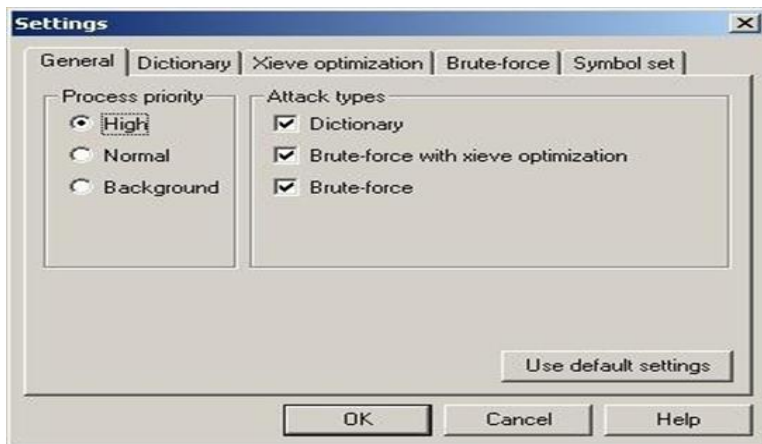
You will find OfficeKey and a MS document in the folder.

Step 2: Open the Office Key – Password Recovery tool

Step 3: Press the “Recover” button in the upper left corner, or select File Recover

Step 4: Choose the password protected MS Office File you have saved to the Desktop.

Step 5: After running the first password auditing session, check to see if Office key has cracked the password. If the password has not been cracked press the Settings button on the upper tool bar.



Step 6: Once in the settings menu you will be able to modify the search parameters and customize a more targeted search.



Step 7: Repeat steps 3 and 4 until the password has been cracked and opens the MS Office File.

Step 8: Write down the contents of the MS word document and the password into your lab report and submit it to your TA.

TASK 2 – PASSWORD AUDITING (WINDOWS PLATFORM)

The purpose of this task is to familiarize you with act of password cracking/recovery. Password cracking software uses a variety of approaches, including intelligent guessing, dictionary attacks and automation that tries every possible combination of characters. Given enough time the automated method can crack any password, but more effective passwords will last months before breaking.

Thus, revealing the passwords is simply a mater of CPU time and dictionary size

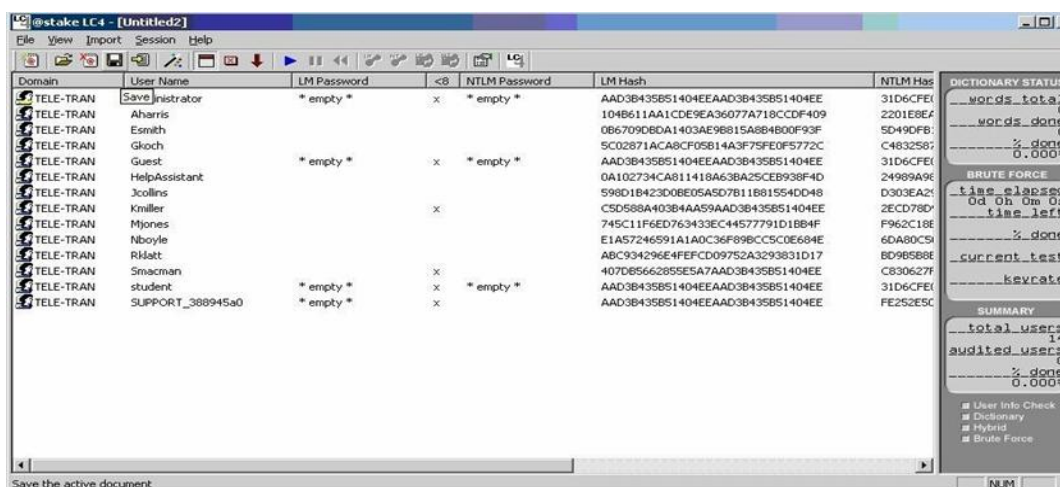
1. You obtain a dictionary file, which is no more than a flat file list of words.
2. These words are fed through any number of programs that encrypt each word. Such encryption conforms to the DES standard.
3. Each resulting encrypted word is compared with the target password. If a match occurs, there is better than a 90% chance that the password was cracked.

Step 1: Go to Lab1 folder, and open LC4 to audit the passwords on your Windows system.

Select File: New session

Select Import: Import from PWDUMP File.

Select the “Passwords” file that has been provided to you.



OBJECTIVES

This password file has been retrieved from a system that we must gain access to. To do this you must crack as many passwords as possible as quickly as possible. We have captured the user names and encrypted passwords for ten users. The user names follow a standard pattern of first initial and last name, but the passwords have no set standards. We do know that users of this system are encouraged to add numbers and other characters to the words they chose for passwords.

To aid you in cracking these passwords we have managed to collect some basic information about the users. This personal information may help you target your searches as to what the user's password may be.

Kmiller	Ken Miller is an avid fly fisher and his record number of catches is just under 30
Smacman	Steven MacMan has a fiancé who's name is 4 letters long and starts with a "K"
Gkoch	Gina Koch grew up with her German grandmother, who used to call her 'Little Precious' *
Mjones	Matt Jones was born in 1979. He compares himself to a Shakespearean character who was born via C section
Tgriffin	Tim Griffin loves funky '70's and '80s music. And songs about 'Love'
Rklatt	Ryan Klatt is a big Star Trek fan and has most likely chosen an obscure reference for his password *
Nboyle	Nancy Boyle is an a fan of the books of British writer Douglas Adams
Esmith	Edward Smith was very close to his grandfather who died in 1968. We know his grandfather's name was a less common name startingwith 'L'
Jcollins	Jim Collins keeps a copy of the book "The Prince" *
Hharris	Alan Harris has a wife named Sue and a daughter named Megan Alan was married on May 3 rd . His daughter was born on August 6 th

Step 2: Select Session options

Step 3: Select session begin “Audit” or press the blue play button on the upper toolbar to start the password search.

Step 4: After the first search has run check your progress. Have some of the passwords been cracked all the way though or have some only been partially cracked. Use what you’ve learned from this first search to target your next few searches. You will need to search the internet and use the information you have been given about each user to find words they may have used as their password.

Step 5: Add words to your wordlist.

Session: Session options.

Press the ‘Dictionary List’ button in the dictionary crack section. Here you can edit your current word list and add words by selecting the ‘EDIT’ button and entering each word on a new line. You can also add multiple dictionaries and wordlist.

Step 6: You may choose to conduct dictionary attacks with other wordlists. You can find additional wordlist to sue here:



Step 7: Continue searching for possible passwords during the remainder of the lab. Repeating steps 3 and 4 each time you modify your search.

Step 8: Once you have cracked all the passwords in the file, write them down in your lab reporter once the lab time has ended, submit the passwords you were able to crack.

RESULT:

Thus the experiment for eavesdropping, dictionary attacks, MITM attacks was done successfully.

Ex no: 07	Sniff Traffic using ARP Poisoning
Date :	

AIM

Perform an experiment to sniff traffic using ARP Poisoning.

DESCRIPTION

ARP is the acronym for Address Resolution Protocol. It is used to convert IP address to physical addresses [MAC address] on a switch. The host sends an ARP broadcast on the network, and the recipient computer responds with its physical address [MAC address]. The resolved IP/MAC address is then used to communicate. **ARP poisoning is sending fake MAC addresses to the switch so that it can associate the fake MAC addresses with the IP address of a genuine computer on a network and hijack the traffic.**

ARP POISONING COUNTERMEASURES

Static ARP Entries: These can be defined in the local ARP cache and the switch configured to ignore all auto ARP reply packets. The disadvantage of this method is, it's difficult to maintain on large networks. IP/MAC address mapping has to be distributed to all the computers on the network. **ARP poisoning detection software:** these systems can be used to cross check the IP/MAC address resolution and certify them if they are authenticated. Uncertified IP/MAC address resolutions can then be blocked.

Operating System Security: This measure is dependent on the operating system been used. The following are the basic techniques used by various operating systems.

- **Linux:** These work by ignoring unsolicited ARP reply packets.
- **Microsoft Windows:** The ARP cache behaviour can be configured via the registry. The following list includes some of the software that can be used to protect networks against sniffing.
- **AntiARP:** provides protection against both passive and active sniffing.
- **Agnitum Outpost Firewall:** provides protection against passive sniffing.
- **XArp:** provides protection against both passive and active sniffing.
- **MacOS:** ArpGuard can be used to provide protection. It protects against both active and passive sniffing.

- Computers communicate using networks. These networks could be on a local area network LAN or exposed to the internet. **Network Sniffers are programs that capture low level package data that is transmitted over a network.** An attacker can analyze this information to discover valuable information such as user ids and passwords.
- In this article, we will introduce you to common network sniffing techniques and tools used to sniff networks.

What is network sniffing?

Computers communicate by broadcasting messages on a network using IP addresses. Once a message has been sent on a network, the recipient computer with the matching IP address responds with its MAC address.

Network sniffing is the process of intercepting data packets sent over a network

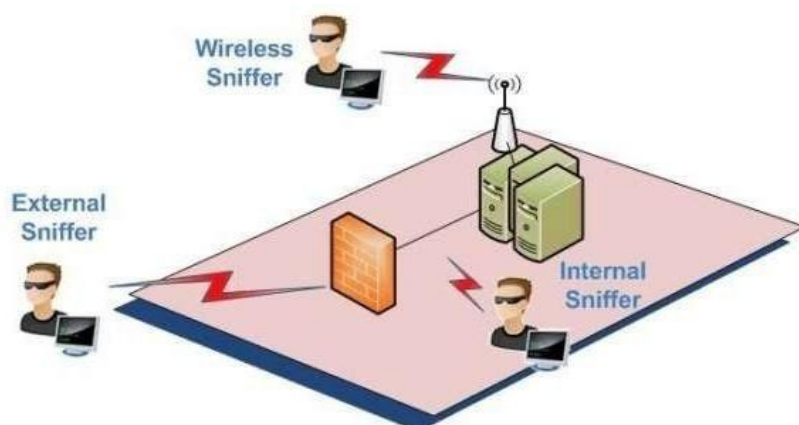
This can be done by the specialized software program or hardware equipment. Sniffing can be used to:

- Capture sensitive data such as login credentials.
- Eavesdrop on chat messages
- Capture files have been transmitted over a network.

The following are protocols that are vulnerable to sniffing

1. Telnet
2. Rlogin
3. HTTP
4. SMTP
5. NNTP
6. POP
7. FTP
8. IMAP

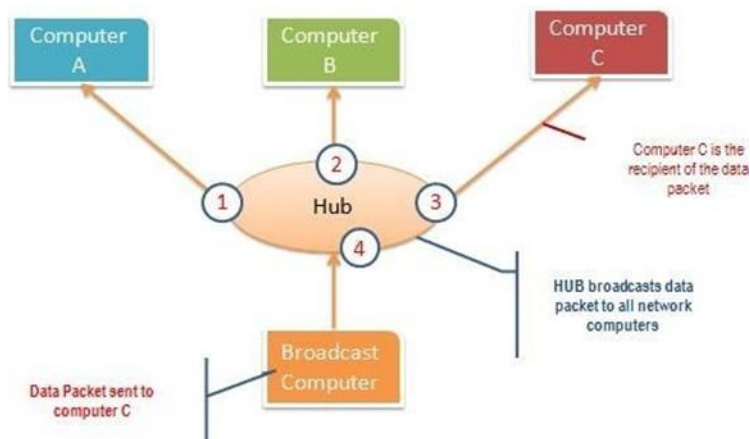
The above protocols are vulnerable if login details are sent in plain text.



Passive and Active Sniffing

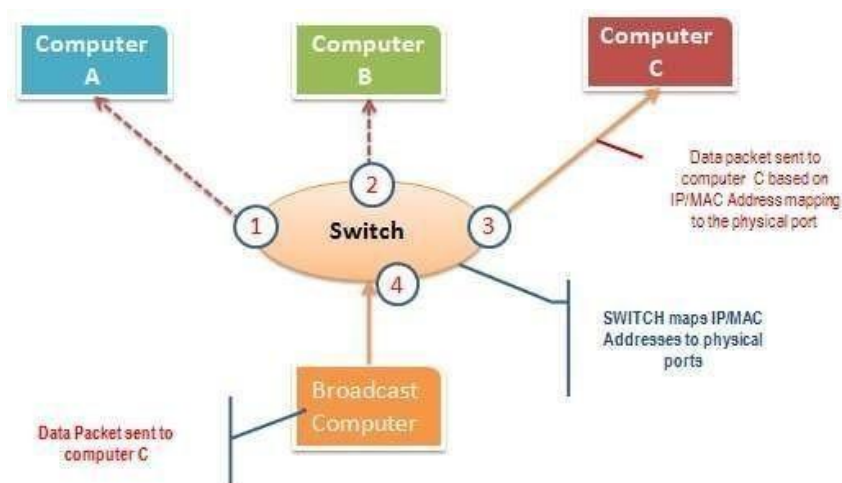
Before we look at passive and active sniffing, let's look at two major devices used to network computers, hubs and switches. **A hub works by sending broadcast messages to all output ports on it except the one that has sent the broadcast.** The recipient computer responds to the broadcast message if the IP address matches. This means when using a hub, all the computers on a network can see the broadcast message. It operates at the physical layer of the OSI Model.

The diagram below illustrates how the hub works.



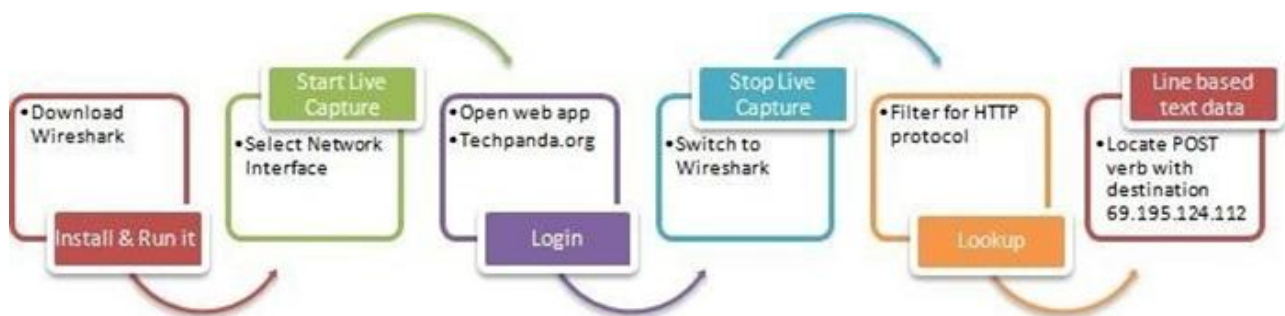
A switch works differently: it maps IP/MAC addresses to physical ports on it. Broadcast messages are sent to the physical ports that match the IP/MAC address configurations for the recipient computer. This means broadcast messages are only seen by the recipient computer. Switches operate at the data link layer and network layer.

The diagram below illustrates how the switch works.



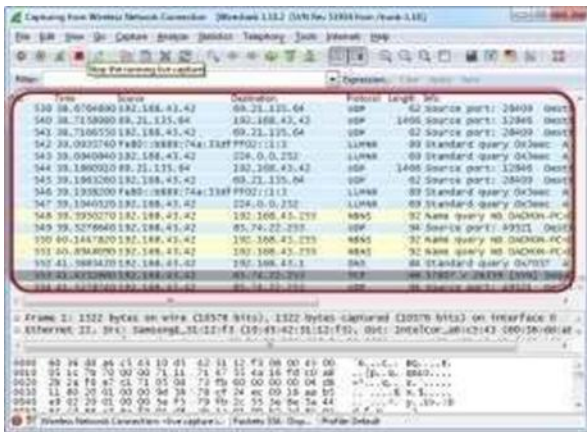
Passive sniffing is intercepting packages transmitted over a network that uses a hub. It is called passive sniffing because it is difficult to detect. It is also easy to perform as the hub send broadcast messages to all the computers on the network.

Active sniffing is intercepting packages transmitted over a network that uses a switch. There are two main methods used to sniff switch linked networks , ARP poisoning, and MAC flooding sniffing the network using Wireshark. The illustration below shows you the steps that you will carry out to complete this exercise without confusion.

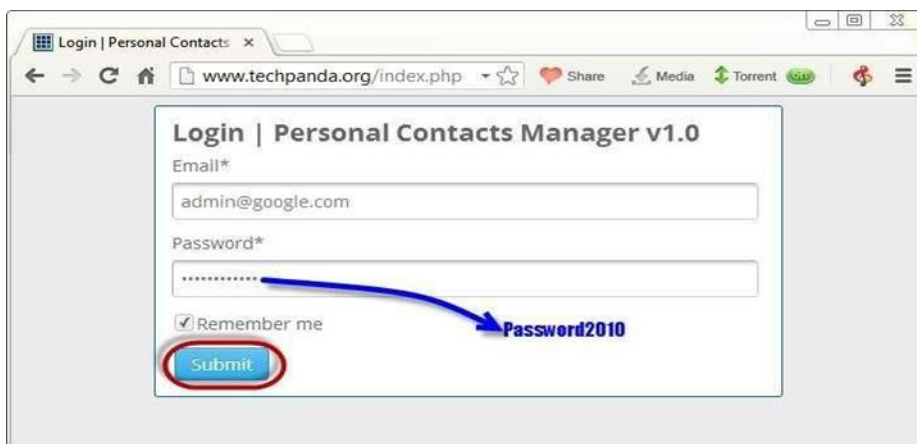


Download wireshark from the link <http://www.wireshark.org/download.html>





Open your web browser and type in <http://www.techpanda.org/>



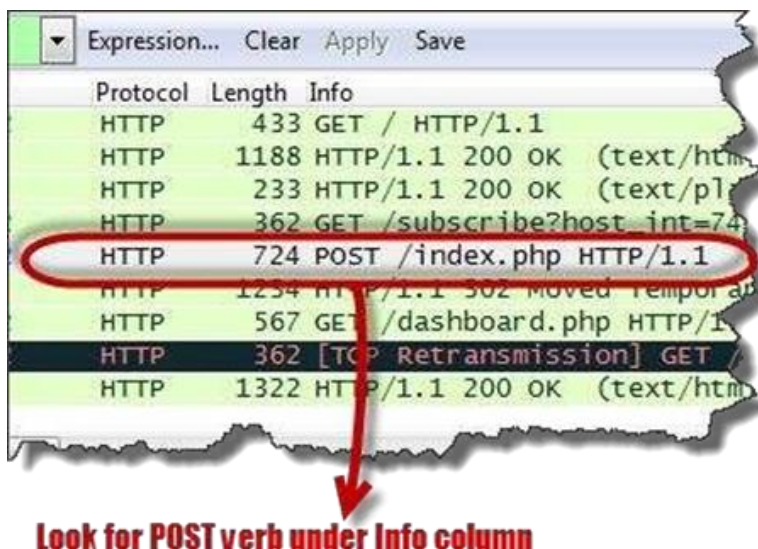
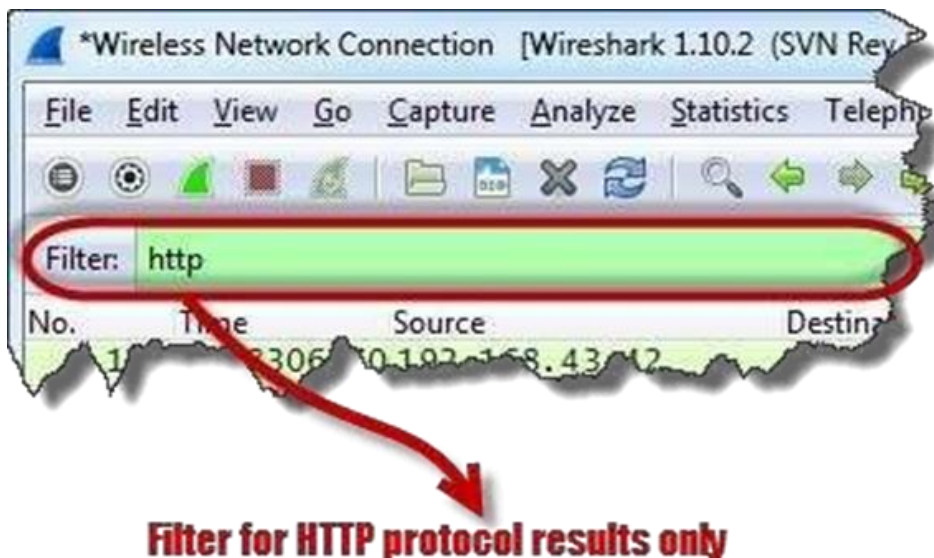
The login email is admin@google.com and the password is **Password2010**

Then click on submit button



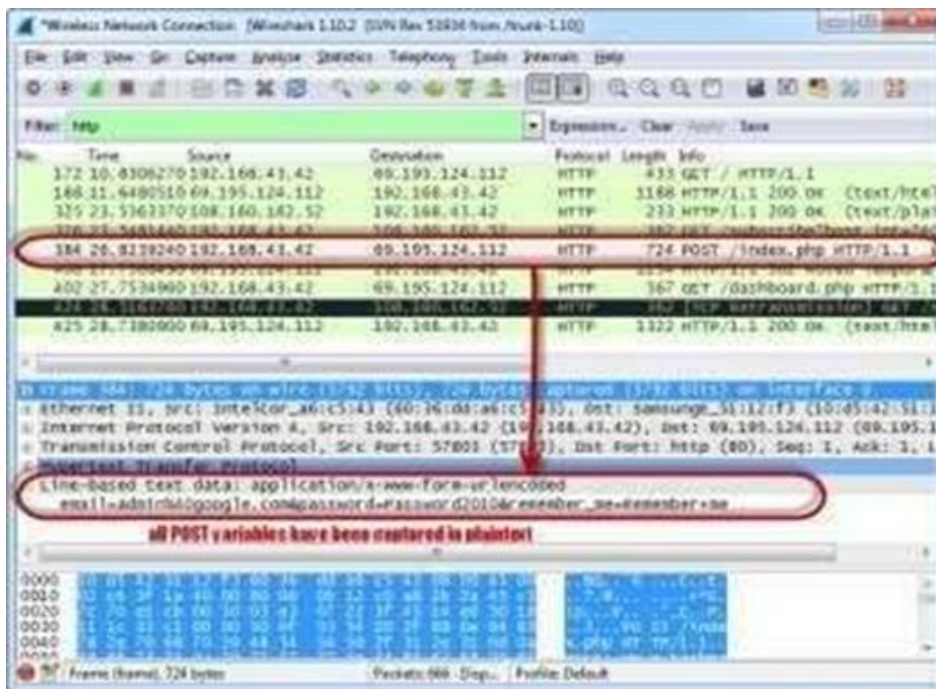


Filter for HTTP protocol results only using the filter text box.



Locate the info column and look for entries with the HTTP verb POST and click on it

Just below the log entries, there is a panel with a summary of captured data. Look for the summary that says line based text data: application/x-www-form-urlencoded



You should be able to view the plaintext values of all the POST variables submitted to the server via HTTP protocol.

RESULT

Thus the experiment to Sniff traffic using ARP poisoning was performed.

Ex no: 08

Date :

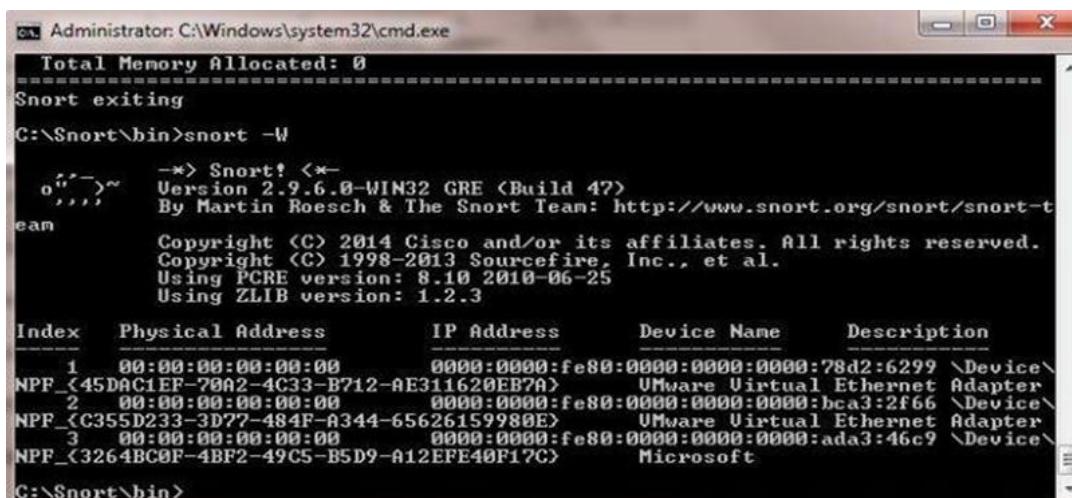
Demonstration of Intrusion Detection System(IDS)

AIM

To demonstrate Intrusion Detection System (IDS) using snort software tool.

STEPS ON CONFIGURING AND INTRUSION DETECTION

1. Download Snort from the Snort.org website (<http://www.snort.org/snort-downloads>)
2. Download Rules(<https://www.snort.org/snort-rules>). You must register to get the rules.
3. Double click on the .exe to install snort. This will install snort in the “C:\Snort” folder. It is important to have WinPcap(<https://www.winpcap.org/install/>) installed.
4. Extract the Rules file. You will need WinRAR for the .gz file.
5. Copy all files from the “rules” folder of the extracted folder of the extracted folder. Now paste the rules into “C:\Snort\rules” folder.
6. Copy “snort.conf” file from the “etc” folder of the extracted folder. You must paste it into “C:\Snort\etc” folder. Overwrite any existing file. Remember if you modify your snort.conf file and download a new file, you must modify it for Snort to work.
7. Open a command prompt (cmd.exe) and navigate to folder “C:\Snort\bin” folder. (at the Prompt, type cd\snort\bin)
8. To start (execute) snort in sniffer mode use following command:
 - a. Snort -dev-I 3
9. To check the interface list, use following command
 - a. Snort-W



```
Administrator: C:\Windows\system32\cmd.exe
Total Memory Allocated: 0
=====
Snort exiting
C:\Snort\bin>snort -W

o"~>~
"'"
ean

-> Snort! <*-
Version 2.9.6.0-WIN32 GRE (Build 47)
By Martin Roesch & The Snort Team: http://www.snort.org/snort/snort-t
ean
Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using PCRE version: 8.10 2010-06-25
Using ZLIB version: 1.2.3

Index   Physical Address      IP Address             Device Name            Description
-----
1       00:00:00:00:00:00      0000:0000:fe80:0000:0000:0000:78d2:6299 \Device\
NPF_{45DAC1EF-70A2-4C33-B712-AE311620EB7A} VMware Virtual Ethernet Adapter
2       00:00:00:00:00:00      0000:0000:fe80:0000:0000:0000:bca3:2f66 \Device\
NPF_{C355D233-3D77-484F-A344-65626159980E} VMware Virtual Ethernet Adapter
3       00:00:00:00:00:00      0000:0000:fe80:0000:0000:0000:ada3:46c9 \Device\
NPF_{3264BC0F-4BF2-49C5-B5D9-A12EFE40F17C} Microsoft
C:\Snort\bin>
```


You can tell which interface to use by looking at the Index number and finding Microsoft. As you can see in the above example, the other interfaces are for VMWare.

To run snort in IDS mode, you will need to configure the file “snort.conf” according to your network environment.

To specify the network address that you want to protect in snort.conf file, look for the following line. `var HOME_NET 192.168.1.0/24` (You will normally see any here)

You may also want to set the addresses of `DNS_SERVERS`, if you have some on your network.

Example:

example snort

Change the `RULE_PATH` variable to the path of rules folder.
`var RULE_PATH`
`c:\snort\rules`

path to rules

Change the path of all library files with the name and path on your system. and you must change the path of `snort_dynamicpreprocessor` variable.
`C:\Snort\lib\snort_dynamicpreprocessor`

You need to do this to all library files in the “C:\Snort\lib” folder. The old path might be: “/usr/local/lib/...”. you will need to replace that path with your system path. Using `C:\Snort\lib` Change the path of the “dynamicengine” variable value in the “snort.conf” file..

Example

dynamicengine `C:\Snort\lib\snort_dynamicengine\sfe_engine.dll`

Add the paths for “include classification.config” and “include reference.config” files.
`include c:\snort\etc\classification.config`
`include c:\snort\etc\reference.config`

Remove the comment (#) on the line to allow ICMP rules, if it is commented with a #.
`include $RULE_PATH/icmp.rules`

You can also remove the comment of ICMP-info rules comment, if it is commented. `include $RULE_PATH/icmp-info.rules`

To add log files to store alerts generated by snort, search for the “output log” test in snort.conf and add the following line:

output alert_fast: snort-alerts.ids

Comment (add a #) the whitelist \$WHITE_LIST_PATH/white_list.rules and the blacklist

Change the nested_ip inner , \ to
nested_ip inner #, \Comment out (#)
following lines:

```
#preprocessor normalize_ip4
#preprocessor normalize_tcp:
ips ecn stream#preprocessor
normalize_icmp4
#preprocessor normalize_ip6
#preprocessor normalize_icmp6
```

Save the “snort.conf” file.

To start snort in IDS mode, run the following command:

```
snort -c c:\snort\etc\snort.conf -l
c:\snort\log -i 3(Note: 3 is used
for my interface card)
```

If a log is created, select the appropriate program to open it. You can use WordPard or NotePad++ to read the file.

To generate Log files in ASCII mode, you can use following command while running snort in IDS mode:

```
snort -A console -i3 -c c:\Snort\etc\snort.conf -l c:\Snort\log -K ascii
```

Scan the computer that is running snort from another computer by using PING or NMap (ZenMap).

After scanning or during the scan you can check the snort-alerts.ids file in the log folder to insure it is logging properly. You will see IP address folders appear.

Snort monitoring traffic –

```
Administrator: C:\Windows\system32\cmd.exe - snort -A console -i3 -c c:\Snort\etc\snort.conf -l c:\Snort\var\log
Rules Engine: SF_SNORT_DETECTION_ENGINE Version 2.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Commencing packet processing (pid 2164)
03/29-23:53:16.033913 [**] [120:3:1] <http_inspect> NO CONTENT-LENGTH OR TRANSF
ER-ENCODING IN HTTP RESPONSE [**] [Classification: Unknown Traffic] [Priority: 3]
1 <TCP> 192.168.1.1:80 -> 192.168.1.20:56506
03/29-23:53:16.035372 [**] [120:3:1] <http_inspect> NO CONTENT-LENGTH OR TRANSF
ER-ENCODING IN HTTP RESPONSE [**] [Classification: Unknown Traffic] [Priority: 3]
1 <TCP> 192.168.1.1:80 -> 192.168.1.20:56507
03/29-23:53:16.036479 [**] [120:3:1] <http_inspect> NO CONTENT-LENGTH OR TRANSF
ER-ENCODING IN HTTP RESPONSE [**] [Classification: Unknown Traffic] [Priority: 3]
1 <TCP> 192.168.1.1:80 -> 192.168.1.20:56508
03/29-23:53:16.037093 [**] [120:3:1] <http_inspect> NO CONTENT-LENGTH OR TRANSF
ER-ENCODING IN HTTP RESPONSE [**] [Classification: Unknown Traffic] [Priority: 3]
1 <TCP> 192.168.1.1:80 -> 192.168.1.20:56509
03/29-23:53:16.142921 [**] [120:3:1] <http_inspect> NO CONTENT-LENGTH OR TRANSF
ER-ENCODING IN HTTP RESPONSE [**] [Classification: Unknown Traffic] [Priority: 3]
1 <TCP> 192.168.1.1:80 -> 192.168.1.20:302
03/29-23:53:16.194409 [**] [120:3:1] <http_inspect> NO CONTENT-LENGTH OR TRANSF
ER-ENCODING IN HTTP RESPONSE [**] [Classification: Unknown Traffic] [Priority: 3]
1 <TCP> 192.168.1.1:80 -> 192.168.1.20:56510
03/29-23:53:16.677078 [**] [120:3:1] <http_inspect> NO CONTENT-LENGTH OR TRANSF
ER-ENCODING IN HTTP RESPONSE [**] [Classification: Unknown Traffic] [Priority: 3]
1 <TCP> 192.168.1.1:80 -> 192.168.1.20:56512
03/29-23:53:16.808301 [**] [120:3:1] <http_inspect> NO CONTENT-LENGTH OR TRANSF
ER-ENCODING IN HTTP RESPONSE [**] [Classification: Unknown Traffic] [Priority: 3]
1 <TCP> 192.168.1.1:80 -> 192.168.1.20:56513
03/29-23:53:16.944237 [**] [120:3:1] <http_inspect> NO CONTENT-LENGTH OR TRANSF
ER-ENCODING IN HTTP RESPONSE [**] [Classification: Unknown Traffic] [Priority: 3]
1 <TCP> 192.168.1.1:80 -> 192.168.1.20:56514
03/29-23:53:16.948012 [**] [120:3:1] <http_inspect> NO CONTENT-LENGTH OR TRANSF
ER-ENCODING IN HTTP RESPONSE [**] [Classification: Unknown Traffic] [Priority: 3]
1 <TCP> 192.168.1.1:80 -> 192.168.1.20:56515
03/29-23:53:16.953992 [**] [120:3:1] <http_inspect> NO CONTENT-LENGTH OR TRANSF
ER-ENCODING IN HTTP RESPONSE [**] [Classification: Unknown Traffic] [Priority: 3]
1 <TCP> 192.168.1.1:80 -> 192.168.1.20:56516
03/29-23:53:16.967744 [**] [120:3:1] <http_inspect> NO CONTENT-LENGTH OR TRANSF
ER-ENCODING IN HTTP RESPONSE [**] [Classification: Unknown Traffic] [Priority: 3]
1 <TCP> 192.168.1.1:80 -> 192.168.1.20:56517
03/29-23:53:16.982649 [**] [120:3:1] <http_inspect> NO CONTENT-LENGTH OR TRANSF
ER-ENCODING IN HTTP RESPONSE [**] [Classification: Unknown Traffic] [Priority: 3]
1 <TCP> 192.168.1.1:80 -> 192.168.1.20:56518
```

RESULT

Thus the Intrusion Detection System(IDS) has been demonstrated by using the Open Source Snort Intrusion Detection Tool.

Ex no: 09	Network Monitoring Tools
Date :	

AIM

To explore about Network monitoring tools.

Networking monitoring is an essential part of network management. It involves using various tools to monitor a system network and determine slowness and weak connections, among other issues. Knowing more about these tools can help you understand them better and use the right ones that suit your requirements.

What are network monitoring tools?

Network monitoring tools are software that you can use to evaluate network connections. These software programs can help you monitor a network connection and identify network issues, which may include failing network components, slow connection speed, network outage or unidentifiable connections.

Network Monitoring Tools

Here are eight monitoring tools along with their descriptions and features:

1. SolarWinds Network Performance Monitor

- a. SolarWinds Network Performance Monitor is a multi vendor monitoring tool. IT allows users to monitor multiple vendors' networks at the same time. It also provides network insights for thorough visibility into the health of the networks.

2. Anvik

- a. Auvik is a network monitoring and management tool. It offers a quick implementation process that helps users to set up the tool easily. It also has a clean user interface that makes it easy to navigate and use. The tool provides in depth network visibility that enables faster troubleshooting for network issues. Users can automate network visibility using Auvik. It provides real time updates on network issues and configuration changes.

3. Datadog Network Monitoring

- a. Datadog Network Monitoring offers services for on premises devices and cloud networks. A highlighting feature of this tool is the visual isolations. It offers various graphical representation of all.

4. Paessler PRTG Network Monitor

- a. Paessler's network connection monitoring tool provides a clean user interface and network visibility on multiple devices. Users can track the health of different connection types like LAN, WAN, servers, websites, applications and services.

5. ManageEngine OpManager

- a. ManageEngine OpManager is a good network monitoring and managing tool for users that prefer in depth view of network health and issues. This tool provides over 2000 network performance monitors that allow users to track and monitor their connections and perform detailed analyses on issues.

6. Domotz

- a. Domotz is an expansive tool that provides a list of features for monitoring network connections. It allows users to customise their network monitoring preferences. User can write scripts to retrieve the data they wish to evaluate. It also allows connection to open ports on remote devices while ensuring network security. Users can also scan and monitor network connections globally. Domotz also allows to backup and restore network configuration for switches, firewalls and access points and alerts when there is a change in the configuration.

7. Checkmk

- a. Checkmk is a tool that allows users to automate it completely. You can customize its operations and enable it to perform tasks automatically. It also identifies network and security components without the user requiring manual set up. For example, the tool can identify a firewall even if the user has not set it up. Its Agent Bakery feature enables users to manage network connections. The tool also includes over 2000 plug ins for enhancing network monitoring.

8. Progress Whatsup Gold

- a. Progress Whatsup Gold is a basic network monitoring software. It provides a minimal user interface with essential features like device monitoring, application monitoring, analysing network traffic and managing configurations. The tool allows users to monitor cloud devices, inspect suspicious connections, automate configuration backups and identify, and resolve bandwidth issues.

OTHER TOOLS FOR NETWORK MONITORING

Here are three additional tools for network monitoring:

1. **Forta Intermapper:** This tool enables users to monitor network connections using network maps, allowing them to get a holistic view of all the connections. It also provides various colour codes for different network status, along with real time notifications through text, email and sound.
2. **Nagios Core:** Nagios Core is a monitoring engine that works as the primary application for all.
3. **Zabbix:** Zabbix provides a thorough network monitoring solution with features like server monitoring, cloud monitoring, application monitoring and service monitoring. The tool also includes features like metric collection, business monitoring and root cause analyses of network issues, and allows users to establish a threshold for connection anomalies.

Tips to Choose a Network Monitoring and Management Tool

Here are some useful tips that you can consider while selecting a tool for network monitoring:

Understand the requirements

Understanding why you require network monitoring software is important in the process. Define what feature you want and for what purpose. This can help you identify the right tool for your use. It may also help you choose the correct subscription plan on paid tools.

Browse multiple tools

Once you identify the requirements, consider browsing multiple tools. Visit the websites for the tools and look for the features you require. Spend time studying the features and understand how they can be useful to your requirements. You can also identify a few tools and compare their features to each other.

Consider the budget

Some tools may be free to use, while some may require you to purchase a subscription plan. Paid tools typically offer a free trial period of up to 30 days. Once you identify which tool you may like to use, see if it is free or requires payment. If it is a paid tool, try exploring its features and efficiency during the trial period. Consider keeping a backup tool in case the tool you choose does not fit your usage.

RESULT

Thus the network monitoring tools were explored.

Ex no: 10	Study to configure Firewall, VPN
Date :	

AIM

To study the features of firewall in providing network security and to set Firewall security in windows.

Firewall in Windows 7

Windows 7 comes with two firewalls that work together. One is the **Windows Firewall**, and the other is **Windows Firewalls with Advanced Security (WFAS)**. The main difference between them is the complexity of the rule's configuration based on protocols, ports, address and authentication. By default, both firewalls come with predefined set of rules that allows us to utilize network resources. This includes things like browsing the web-, receiving, e-mails, etc. Other standard firewall exceptions are File and Printer Sharing, Network Discovery, Performance Logs and Alerts, Remote Administration, Windows Remote Management, Remote Assistance, Remote Desktop, Windows Media player Network Sharing Service

With firewall in Windows 7 we can configure inbound and outbound rules. By default, all outbound traffic is allowed, and bound response to that traffic are allowed. Inbound traffic initiated from external sources is automatically blocked.

When we first connect to some network, we are prompted to select a network location. This feature is known as Network Location Awareness (NLA), This feature enables us to assign a network profile to the connection based on the location. Different network profiles contain different collections of firewall rules. In Windows 7, different network interfaces. For example, our wired interface can have different profile than our wireless interface. There are three different network profiles available:

Public

Home/Work – private network

Domin – used within a domain

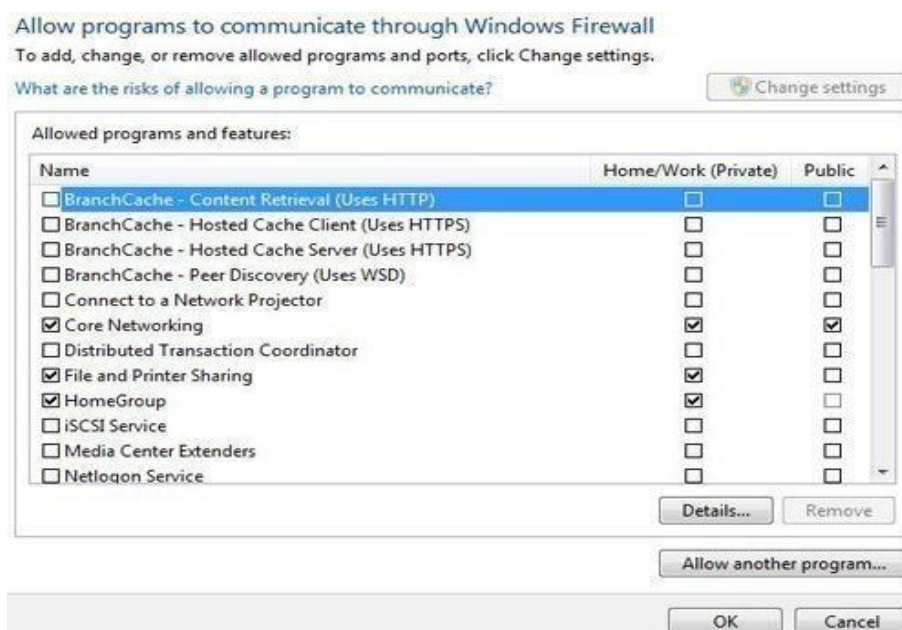
Configuring Windows Firewall

To open Windows Firewall we can go to **Start > Control Panel > Windows**



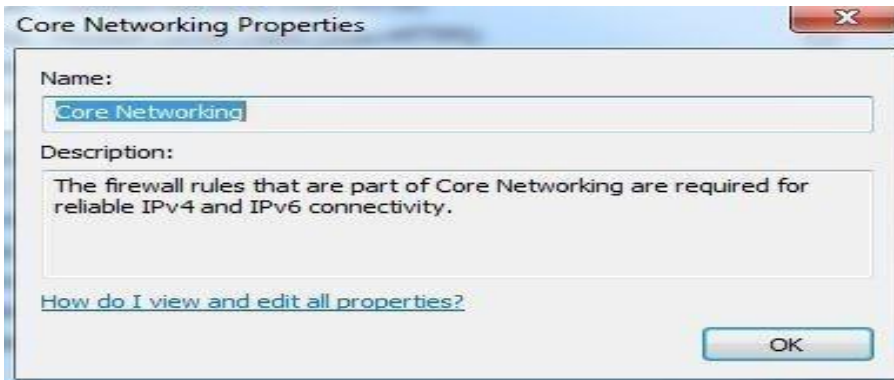
Firewall

By default, Windows Firewall is enabled for both private & public networks. It is also configured to block all connections to programs that are not on the list of allowed programs. To configure exceptions, we can go to the menu on the left and select “Allow a program or feature through Windows Firewall” option.



Exceptions

To change settings in this window we have to click the “Change settings” button. As you can see, here we have a list of predefined programs and features that can be allowed to communicate on private or public networks. For example, notice that the Core Networking feature is allowed on both private and public networks, while the File and Printer Sharing is only allowed on private networks. We can also see the details of the items in the list by selecting it and then clicking the details button.



Details

If we have a program on our computer that is not in this list, we can manually add it by clicking on the “Allow another program” button.

Add a Program

Here we have to browse to the executable of our program and then click the add button. Notice that we can also choose location types on which this program will be allowed to communicate by clicking on the “Network location types” button.

Network Locations

Many applications will automatically configure proper exceptions in Windows Firewall when we run them. For example, if we enable streaming from Media Player, it will automatically configure firewall settings to allow streaming. The same thing is if we enable Remote Desktop feature from the system properties window. By enabling Remote Desktop feature we actually create an exception in Windows Firewall.

Windows Firewall can be turned off completely. To do that we can select the “Turn windows Firewall on or off” option from the menu on the left.

Firewall Customization

Note that we can modify settings for each type of network location (private or public). Interesting thing here is that we can block all incoming connections, including those in the list of allowed programs.

Windows Firewall is actually a Windows service. As you know, services can be stopped and started. If the Windows Firewall service is stopped, the Windows Firewall will not work.

Firewall Service



Service Name	Description	Status	Startup Type
Windows Event Log	This service ...	Started	Automatic
Windows Firewall	Windows Fi...	Started	Automatic
Windows Font Cache S...	Optimizes p...	Started	Automatic (D...
Windows Image Acqui	Provides im		Manual

In our case the service is running. If we stop it, we will get a warning that we should turn on our Windows Firewall.



Warning

Remember that with Windows Firewall we can only configure basic firewall settings, and this is enough for most day-to-day users. However, we can't configure exceptions

based on ports in Windows Firewall any more. For that we have to use Windows Firewall with Advanced Security.

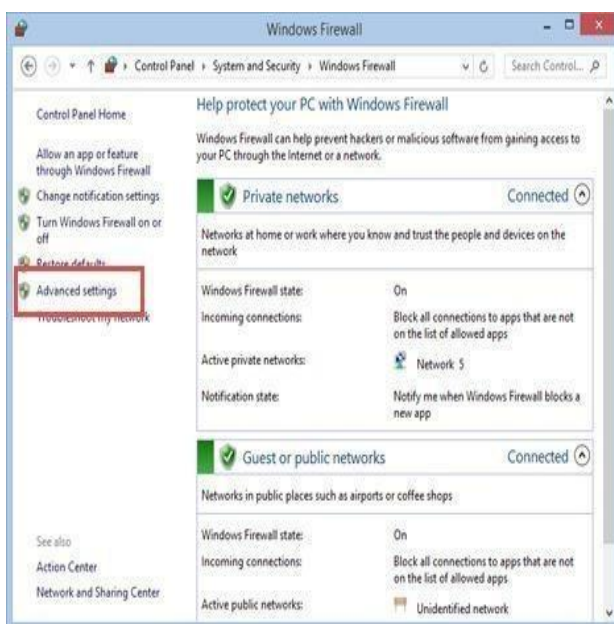
How to Start & Use the Windows Firewall with Advanced Security

The Windows Firewall with Advanced Security is a tool which gives you detailed control over the rules that are applied by the Windows Firewall. You can view all the rules that are used by the Windows Firewall, change their properties, create new rules or disable existing ones. In this tutorial we will share how to open the Windows Firewall with Advanced Security, how to find your way around it and talk about the types of rules that are available and what kind of traffic they filter.

How to Access the Windows Firewall with Advanced Security

You have several alternatives to opening the Windows Firewall with Advanced Security:

One is to open the standard Windows Firewall window, by going to “Control Panel -> System and Security -> Windows Firewall”. Then, click or tap Advanced settings.



In Windows 7, another method is to search for the word firewall in the Start Menu search box and click the “Windows Firewall with Advanced Security” result.

In Windows 8.1, Windows Firewall with Advanced Security is not returned in search results and you need to use the first method shared above for opening it.

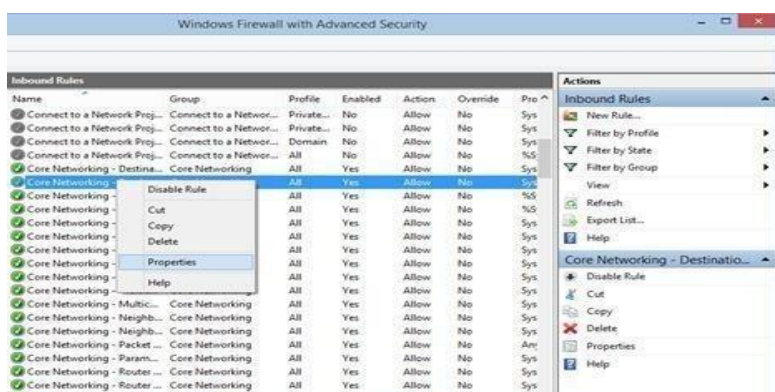


What are the Inbound & Outbound Rules?

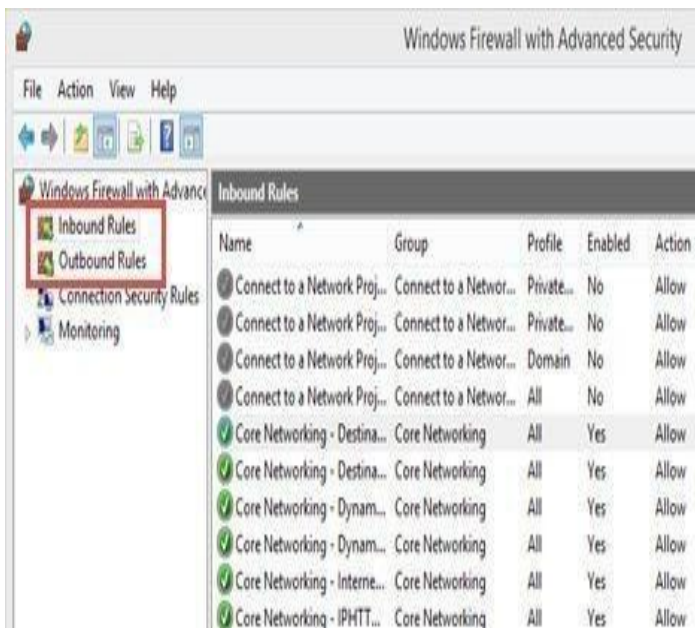
In order to provide the security you need, the Windows Firewall has a standard set of inbound and outbound rules, which are enabled depending on the location of the network you are connected to.

Inbound rules are applied to the traffic that is coming from the network and the Internet to your computer or device. Outbound rules apply to the traffic from your computer to the network or the internet.

These rules can be configured so that they are specific to: computers, users, programs, services, ports or protocols. You can also specify to which type of network adapter or user profile it is applied to.

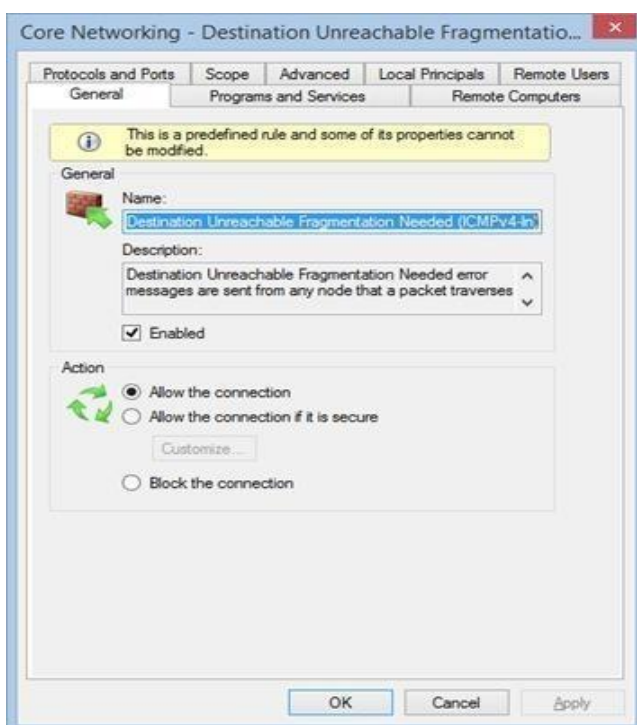


In the Windows Firewall with Advanced Security, you can access all rules and edit their properties. All you have to do is click or tap the appropriate unit in the left side panel.



The rules used by the Windows Firewall can be enabled or disabled. The ones which are enabled or active are marked with a green check box in the Name column. The ones that are disabled are marked with a gray check box.

If you want to know more about a specific rule and learn its properties, right click on it and select Properties or select it and press Properties in the column on right, which lists the actions that are available for your selection.

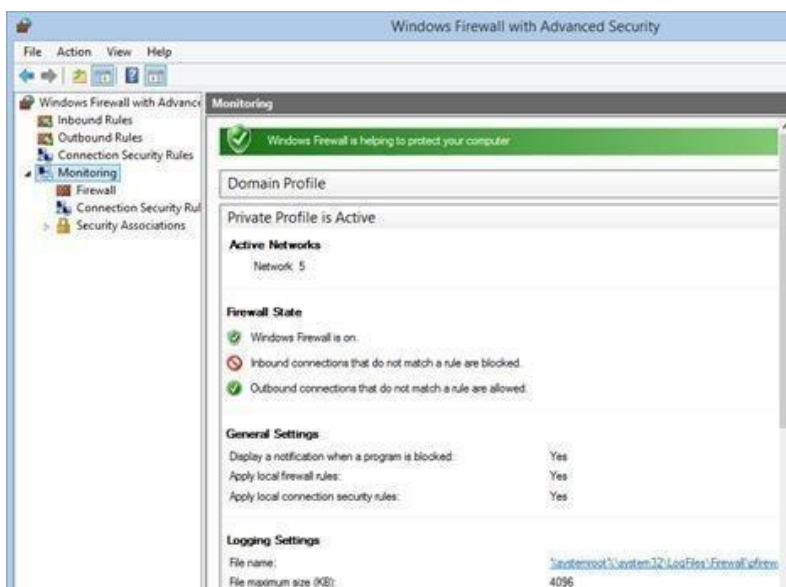


What are the connection security rules?

Connection security rules are used to secure traffic between two computers while it crosses the network. One example would be a rule which defines that connections between two specific computers must be encrypted.

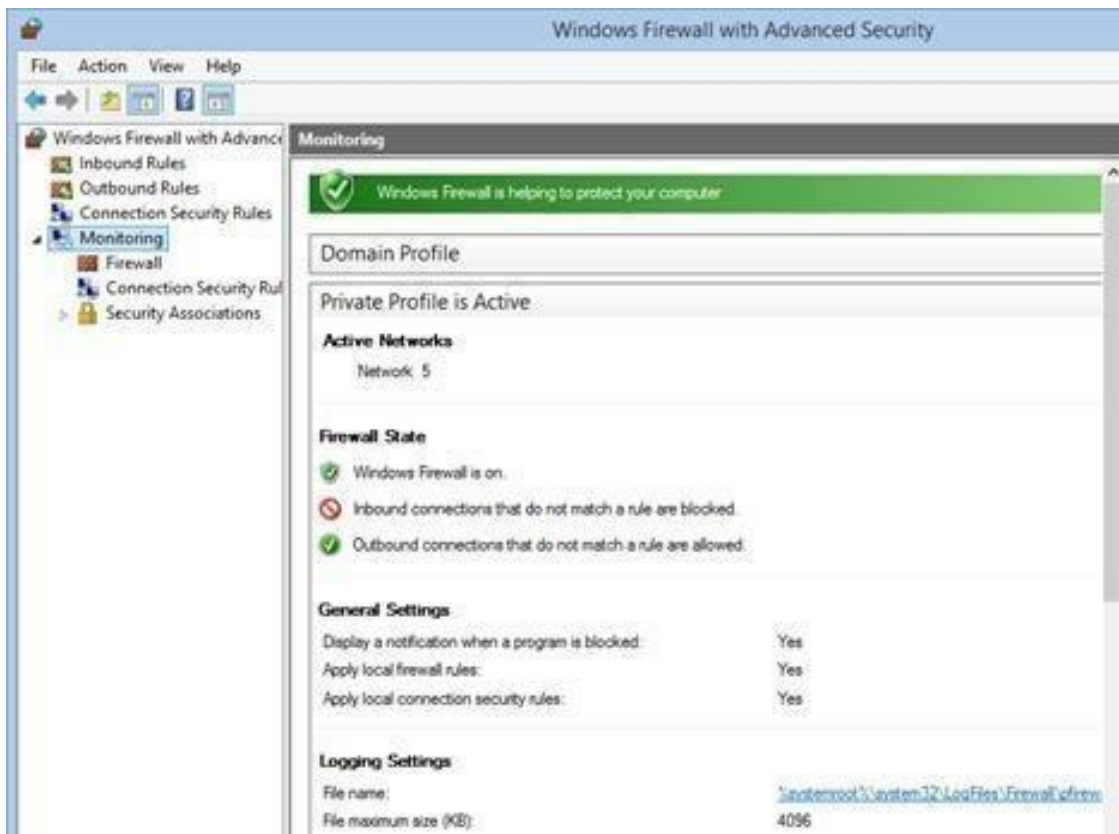
Unlike the inbound or outbound rules, which are applied only to one computer, connection security rules require that both computers have the same rules defined and enabled.

If you want to see if there are any such rules on your computer, click or tap “Connection Security Rules” on the panel on the left. By default, there are no such rules defined on Windows computers and devices. They are generally used in business environments and such rules are set by the network administrator.



What Does the Windows Firewall with Advanced Security Monitor?

The Windows Firewall with Advanced Security includes some monitoring features as well. In the Monitoring section you can find the following information: the firewall rules that are active (both inbound and outbound), the connection security rules that are active and whether there are any active security associations.



You should note that the Monitoring section shows only the active rules for the current network locations.

RESULT

Study of the features of firewall in providing network security and to set Firewall Security in windows.